CP115
Computer Programming 1
Semester 1
Session 2025 / 2026
2 hours 30 minutes

Name: _____ Practicum: _____

KOLEJ MATRIKULASI NEGERI SEMBILAN
NEGERI SEMBILAN MATRICULATION COLLEGE

Do not open this question paper until you are told to do so.

For Examiner's Use		
Question No.	**Mark**	
	Examiner	**KP/KKP**
1 (25)		
2 (13)		
3 (15)		
4 (12)		
5 (15)		
6 (10)		
TOTAL		

This question paper consists of 11 printed pages.

1. Problem solving is a systematic process of finding an effective solution to a specific problem using logical thinking and analysis.

(a) (i) List 2 steps in problem solving.

[*2 marks*]

_____

_____

(ii) Describe **one (1)** of the steps mentioned in Question 1(a)(i).

[*1 mark*]

_____

_____

(b) As a hostel supervisor, **Mr. Amir** develops a system to track student check-ins and check-outs using QR codes. Identify each problem-solving step based on the given scenario.

[*3 marks*]

Scenario	Problem-solving step
i. Once the system is developed, Mr. Amir tests it with multiple QR codes and verifies whether each student's check-in and check-out time are correctly recorded. He fixes any incorrect records.	
ii. Before starting the project, Mr. Amir identifies the inputs (student ID and QR code), the process (record date and time), and the output (student attendance report). This helps him define the problem clearly before solving it.	
iii. During the development process, Mr. Amir writes the program code in Python according to the algorithm he has planned. He ensures that the syntax and logic follow the correct programming rules.	

(c) The Input-Process-Output (IPO) model is a framework that shows how data is received as input, transformed through processing, and produced as output.

(i) Describe the **process** component in the Input-Process-Output (IPO) model.

[*1 mark*]

_____

(ii) For each description below, identify whether it represents **input**, **process**, or **output**.

[*3 marks*]

Description	Answer
i. A parking system scans the car plate number to record entry time.	
ii. A smart washing machine adjusts the water level based on load weight.	
iii. A delivery app shows a notification when the parcel has arrived.	

(d) A swimming pool contractor wants to estimate the cost of waterproof coating for a rectangular pool. The program accepts length, width, and depth, then calculates and displays the surface area and total cost if the cost is RM18.00 per m².

[*4 marks*]

Input	Process	Output

(e) The table below shows how a smart air conditioner operates based on the room temperature detected by an IoT sensor. Based on the table below, write an IPO that controls the smart air conditioner. The program should read the room temperature and display the current temperature, selected mode, and an appropriate message according to the temperature range.

Temperature °C	Air Conditioner Mode	Message
Below 22°C	Off	"Temperature is cool."
22°C – 28°C	Energy-Saving Mode	"Temperature is moderate."
Above 28°C	Cooling Mode	"Temperature is high."

[*6 marks*]

Input	Process	Output

(f) A **stationery shop** recorded payments made by **five customers** who bought notebooks. Each customer paid a different amount. The price of one notebook is **RM2.80**. Write an IPO analysis that repeatedly reads the payment from five customers and then calculates and displays the **number of notebooks** purchased by each customer.

[5 marks]

Input	Process	Output

2. In computer programming, flowchart and pseudocode are commonly used techniques to represent and plan an algorithm.

(a) Explain the meaning of an algorithm.

[*2 marks*]

_____

_____

(b) Differentiate between flowchart and pseudocode.

[*2 marks*]

Flowchart	Pseudocode

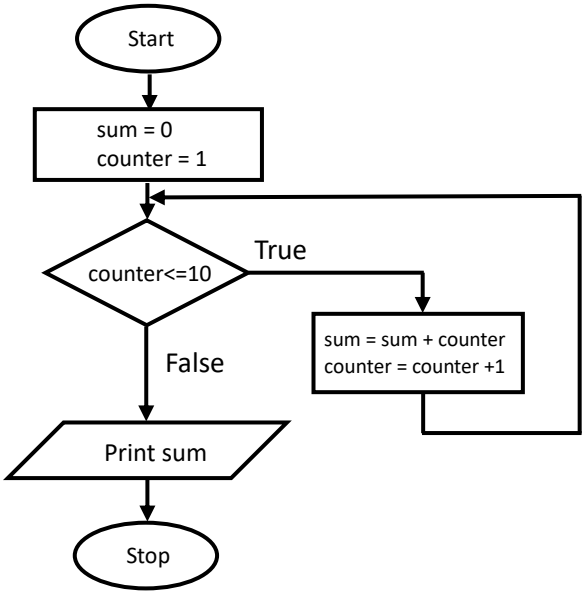
(c) Convert this pseudocode to flowchart.

[*4 marks*]

Pseudocode	Flowchart
Start Input score If score ≥ 50 then Print "Pass" End If Stop	

(d) Convert the following flowchart to pseudocode.

[*5 marks*]



Answer:

3. A programming language is a structured and formal system of communication that enables humans to write instructions in a precise and unambiguous way, allowing computers to perform specific tasks, solve problems, and execute programs efficiently.

(a) Define the following terminology:

(i) Object-Oriented Paradigm [*1 mark*]

_____

_____

(ii) Logic Paradigm [*1 mark*]

_____

_____

(b) Online-based Python platforms let users write and run Python programs through a web browser without installing any software.

State limitation and advantage of online-based Python platforms.

[*2 marks*]

Limitation	Advantage

(c) (i) Identify whether each identifier is valid or invalid.

[2 marks]

Identifier	Valid / Invalid
4thValue	
employee_name	

(ii) Identify the type of error based on the given statement.

[2 marks]

Statement	Type of error

`if score > 50:`
`  print("Fail")`
`# should print "Pass" instead`	

`Divisible by zero`	

(iii) State one (1) reason why **and**, **class**, or **for** cannot be used as identifiers.

[1 mark]

_____

_____

(iv) Identify the most suitable **data type** for each statement.

[2 marks]

Statement	Data Type
Store a student's CGPA (Example: 3.85).	
Store whether a student is eligible for a hostel (True or False).	

(v) Write **Python assignment statements** that accomplish the following:

[2 marks]

Assign $\pi \times r^2$ to variable area.	
Assign value True to variable active.	

(vi) Determine the appropriate **control structure** based on each situation.

[2 marks]

Situation	Type of Control Structure
Determine whether a student passed or failed based on marks.	
Display all student names in a list until the end of the file.	

4. **(a)** Answer the question based on the Python code below.

```
discount = 0
amount = float(input("Enter purchase amount: "))
member = input("Are you a member? (yes/no): ")

if amount > 200:
    discount = amount * 0.15
elif amount >= 100:
    discount = amount * 0.10

if member == "yes":
    discount = discount + 10

print("Total Discount:", discount)
print("Final Price:", amount - discount)
```

Write the output for the following input:

i. amount = 250, member = yes
ii. amount = 120, member = no

[2 marks]

i. _____

ii. _____

(b) Write the output for the following code:

[3 marks]

```
for i in range(2, 10, 3):
    print(i)
print("Done")
```

_____

_____

(c) Define **single selection** and **multiple selection**.

[2 marks]

Structure	Definition
Single selection	
Multiple selection	

(d) Due to the increasing awareness of energy conservation, a smart home system is designed to monitor electricity usage. If the daily electricity consumption exceeds 30 kWh, the system will display "High usage! Please reduce consumption". Otherwise, it will display "Usage is normal".

Write a Python program to perform this task.

[5 marks]

5. Repetition is a programming concept that allows a set of instructions to be executed repeatedly until a specific condition is met.

(a) Give **one (1)** example of Python code segment for counter-controlled and sentinel-controlled.

[*4 marks*]

Structure	**Python code segment**
Counter-controlled	
Sentinel-controlled	

(b) Convert the following pseudocode to Python coding.

[*5 marks*]

Start Read number while number ≠ -1 Print number Read number End while Stop	

(c) In conjunction with an environmental campaign, a school records the amount of plastic waste collected each day for 5 days. Write a Python program using a for in range loop to input and display the total amount of plastic collected.

[*6 marks*]

6. In a Smart City IoT Waste Management System, sensors placed in bins send the waste level percentage (0–100%) to a central monitoring program. The data entry continues until the user enters -1 to indicate there are no more bins to process.

   Construct a Python program that:
   1. Repeatedly inputs waste levels for each bin (sentinel-controlled loop).
   2. Calculates the total, maximum, and average waste level.
   3. Displays the number of bins recorded.
   4. Prints an alert message:
      - "High waste level detected! Schedule collection." if average > 75, or
      - "Waste level under control." otherwise.

   [*10 marks*]

```
Example data and output:

        Enter waste level (0–100) or -1 to stop: 80
        Enter next waste level or -1 to stop: 60
        Enter next waste level or -1 to stop: 90
        Enter next waste level or -1 to stop: 70
        Enter next waste level or -1 to stop: -1

        Number of bins recorded: 4
        Total waste level: 300.0
        Maximum waste level: 90.0
        Average waste level: 75.00%
        High waste level detected! Schedule collection.
```

END OF QUESTION PAPER